2023 Annual Dairy Report Reissued WDR for Existing Milk Cow Dairies

No: R5-2013-0122

Amaral Dairy, 4715 County Road N, Orland, CA 95963

June 6, 2024

→ **The Power of Commitment**

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Amaral Dairy

Physical address of dairy:

4715 County Road N	Orland	Glenn	95963
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 10/13/1990

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

XX44-XX25-XX12-XXX9

B. OPERATORS

Amaral, Jose

Operator name: Amaral, Jose

Telephone no.: (530) 865-7558 (Landline) (Cellular)

4715 County Road N	Orland	CA	95963
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

Amaral, Maria

Operator name: Amaral, Maria

Telephone no.: (530) 865-7558 (Landline) (Cellular)

4715 County Road N	Orland	CA	95963
Mailing Address Number and Street	City	State	Zip Code

C. OWNERS

Amaral, Jose

Legal owner name: Amaral, Jose

Telephone no.: (530) 865-7558
Landline | Cellular

4715 County Road N | Orland | CA | 95963
Mailing Address Number and Street | City | State | Zip Code

This owner is responsible for paying permit fees.

Amaral, Maria

Legal owner name: Amaral, Maria

Telephone no.: (530) 865-7558
Landline | Cellular

4715 County Road N | Orland | CA | 95963
Mailing Address Number and Street | City | State | Zip Code

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	150	51	61	56	33	10
Number under roof	0	0	0	0	0	0
Maximum number	150	51	61	56	33	10
Average number	150	51	61	56	33	10
Avg live weight (lbs)	1,200	1,200	850	850		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 41 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 5,342.55 *tons per reporting period*

Total nitrogen from manure: 66,240.21 *lbs per reporting period*

After ammonia losses (30% loss applied): 46,368.15 *lbs per reporting period*

Total phosphorus from manure: 10,552.33 *lbs per reporting period*

Total potassium from manure: 25,868.43 *lbs per reporting period*

Total salt from manure: 82,354.95 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 174,931 *gallons*

Total nitrogen generated: 952.30 *lbs*

Total phosphorus generated: 113.98 *lbs*

Total potassium generated: 3,117.96 *lbs*

Total salt generated: 11,954.04 *lbs*

	174,931	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	174,931	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal Water	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	Corral solids	750.00 *ton*	As-is	28.6		15,600.00	3,148.50	28,677.70		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	23,400.00	4,722.75	43,016.55	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	23,400.00	4,722.75	43,016.55	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	8	8	2	both	XX44-XX25-XX12-XXX9
Field 2	9	9	2	both	XX44-XX25-XX12-XXX9
Field 3	8	8	2	both	XX44-XX25-XX12-XXX9
Totals for areas that were used for application	25	25	6		
Totals for areas that were not used for application					
Land application area totals	25	25	6		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	80.00 *ton*	Dry-weight		58.3	11,700.00	3,180.00	22,300.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	10.00	97.58	26.52	185.98	817.32

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	160.00 *ton*	Dry-weight		69.3	15,400.00	2,520.00	18,600.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	20.00	189.11	30.95	228.41	957.84

Field 2

Field name: Field 2

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	90.00 *ton*	Dry-weight		58.5	11,400.00	3,300.00	24,000.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	10.00	94.62	27.39	199.20	830.00

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	180.00 *ton*	Dry-weight		73.1	16,300.00	3,000.00	20,800.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	20.00	175.39	32.28	223.81	925.36

Field 3

Field name: Field 3

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	80.00 *ton*	Dry-weight		59.8	14,400.00	3,700.00	27,200.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	10.00	115.78	29.75	218.69	884.40

Field 3

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 8 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	160.00 *ton*	Dry-weight		69.1	15,500.00	2,450.00	18,500.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	20.00	191.58	30.28	228.66	976.44

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/30/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	39.04	10.13	64.13	57.47	32.00 *ton*
Application event totals		39.04	10.13	64.13	57.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	26.79	2.83	80.72	273.07	32,928.00 *gal*
Application event totals		26.79	2.83	80.72	273.07	

Field 1 - 06/20/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	69.28	474,412.00 *gal*
Application event totals		0.00	0.00	0.00	69.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	175.24	1,199,984.00 *gal*
Application event totals		0.00	0.00	0.00	175.24	

Field 1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	146.71	1,004,638.00 *gal*
Application event totals		0.00	0.00	0.00	146.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	114.11	781,385.00 *gal*
Application event totals		0.00	0.00	0.00	114.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	105.96	725,572.00 *gal*
Application event totals		0.00	0.00	0.00	105.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	195.62	1,339,517.00 *gal*
Application event totals		0.00	0.00	0.00	195.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	32.60	223,253.00 *gal*
Application event totals		0.00	0.00	0.00	32.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	195.62	1,339,517.00 *gal*
Application event totals		0.00	0.00	0.00	195.62	

Field 1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	11.30	1.73	44.00	205.10	23,050.00 *gal*
Application event totals		11.30	1.73	44.00	205.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	154.87	1,060,451.00 *gal*
Application event totals		0.00	0.00	0.00	154.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	48.90	334,879.00 *gal*
Application event totals		0.00	0.00	0.00	48.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	195.62	1,339,517.00 *gal*
Application event totals		0.00	0.00	0.00	195.62	

Field 2 - 11/30/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	39.04	10.13	64.13	57.47	36.00 *ton*
Application event totals		39.04	10.13	64.13	57.47	

Field 2 - 11/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	26.79	2.83	80.72	273.07	37,044.00 *gal*
Application event totals		26.79	2.83	80.72	273.07	

Field 2 - 06/20/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	64.33	533,714.00 *gal*
Application event totals		0.00	0.00	0.00	64.33	

06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	162.73	1,349,982.00 *gal*
Application event totals		0.00	0.00	0.00	162.73	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	136.24	1,130,217.00 *gal*
Application event totals		0.00	0.00	0.00	136.24	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	105.96	879,058.00 *gal*
Application event totals		0.00	0.00	0.00	105.96	

Field 2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	98.39	816,268.00 *gal*
Application event totals		0.00	0.00	0.00	98.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	181.65	1,506,957.00 *gal*
Application event totals		0.00	0.00	0.00	181.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	30.27	251,159.00 *gal*
Application event totals		0.00	0.00	0.00	30.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	181.65	1,506,957.00 *gal*
Application event totals		0.00	0.00	0.00	181.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	11.30	1.73	44.00	205.09	25,931.00 *gal*
Application event totals		11.30	1.73	44.00	205.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	143.80	1,193,007.00 *gal*
Application event totals		0.00	0.00	0.00	143.80	

Field 2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	45.41	376,739.00 *gal*
Application event totals		0.00	0.00	0.00	45.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	181.65	1,506,957.00 *gal*
Application event totals		0.00	0.00	0.00	181.65	

Field 3 - 11/30/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Pre-Plant	Corral solids	39.04	10.13	64.13	57.47	32.00 *ton*
Application event totals		39.04	10.13	64.13	57.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	26.79	2.83	80.72	273.07	32,928.00 *gal*
Application event totals		26.79	2.83	80.72	273.07	

Field 3 - 06/20/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage Plant date: 06/20/2023

Field 3 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	64.33	474,412.00 *gal*
Application event totals		0.00	0.00	0.00	64.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	162.73	1,199,984.00 *gal*
Application event totals		0.00	0.00	0.00	162.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	136.24	1,004,638.00 *gal*
Application event totals		0.00	0.00	0.00	136.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	105.96	781,385.00 *gal*
Application event totals		0.00	0.00	0.00	105.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	98.39	725,572.00 *gal*
Application event totals		0.00	0.00	0.00	98.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	181.65	1,339,517.00 *gal*
Application event totals		0.00	0.00	0.00	181.65	

Field 3 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	30.27	223,253.00 *gal*
Application event totals		0.00	0.00	0.00	30.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	181.65	1,339,517.00 *gal*
Application event totals		0.00	0.00	0.00	181.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	11.30	1.73	44.00	205.10	23,050.00 *gal*
Application event totals		11.30	1.73	44.00	205.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	143.80	1,060,451.00 *gal*
Application event totals		0.00	0.00	0.00	143.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	45.41	334,879.00 *gal*
Application event totals		0.00	0.00	0.00	45.41	

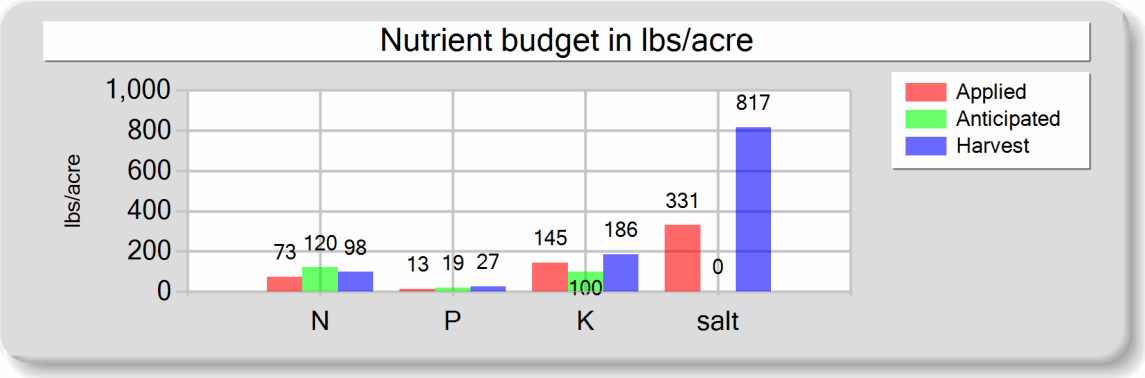
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	181.65	1,339,517.00 *gal*
Application event totals		0.00	0.00	0.00	181.65	

B. NUTRIENT BUDGET

Field 1 - 11/30/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	39.04	10.13	64.13	57.47
Process wastewater	26.79	2.83	80.72	273.07
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	72.83	12.96	144.85	330.54
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	97.58	26.52	185.98	817.32
Nutrient balance	-24.75	-13.56	-41.13	-486.78
Applied to removed ratio	0.75	0.49	0.78	0.40

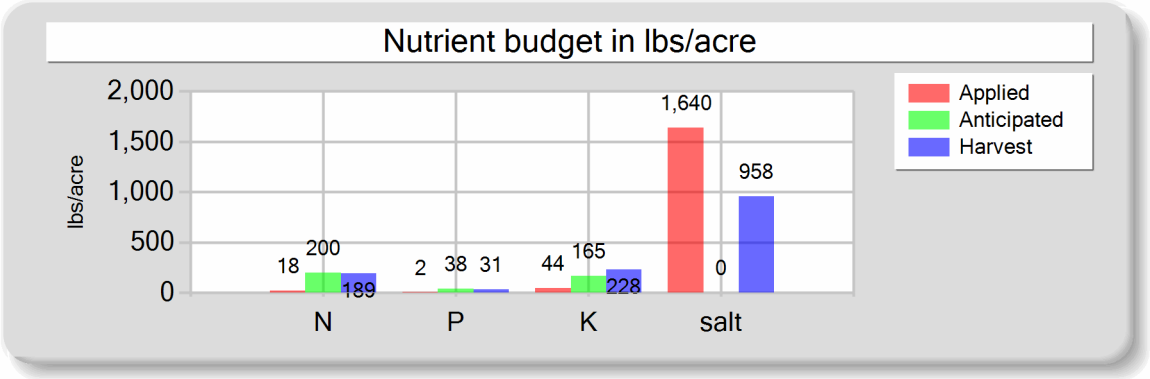
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
32,928.00 *gallons*
1.21 *acre-inches*
0.15 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 1 - 06/20/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	11.30	1.73	44.00	205.10
Fresh water	0.00	0.00	0.00	1,434.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	18.30	1.73	44.00	1,639.64
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	189.11	30.95	228.41	957.84
Nutrient balance	-170.81	-29.22	-184.41	681.80
Applied to removed ratio	0.10	0.06	0.19	1.71

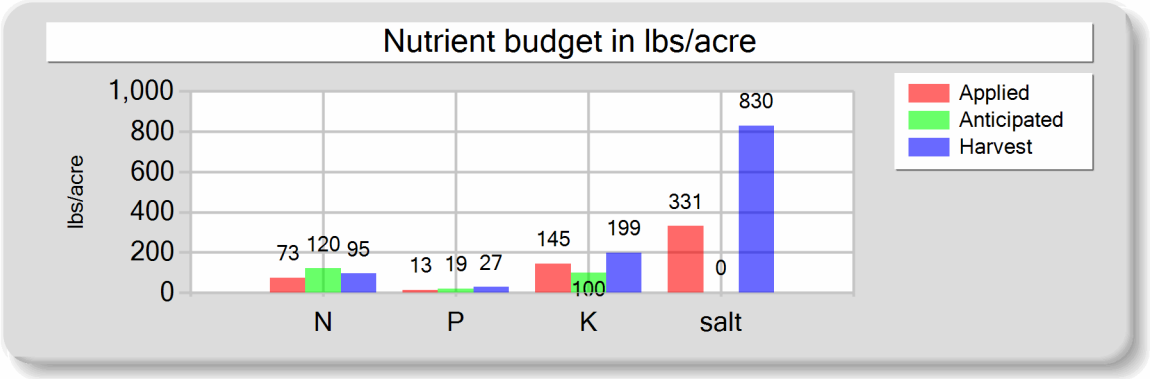
Fresh water applied	
9,823,125.00	*gallons*
361.75	*acre-inches*
45.22	*inches/acre*

Process wastewater applied	
23,050.00	*gallons*
0.85	*acre-inches*
0.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/30/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	39.04	10.13	64.13	57.47
Process wastewater	26.79	2.83	80.72	273.07
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	72.83	12.96	144.85	330.54
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	94.62	27.39	199.20	830.00
Nutrient balance	-21.79	-14.43	-54.35	-499.46
Applied to removed ratio	0.77	0.47	0.73	0.40

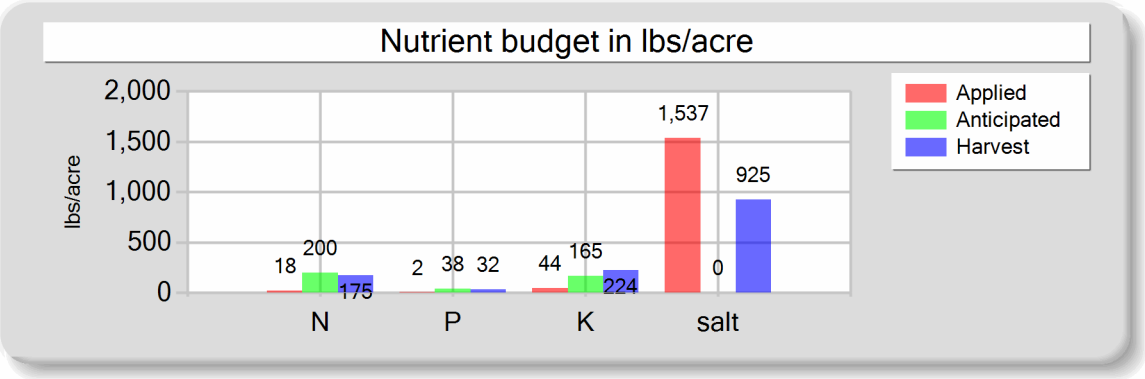
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
37,044.00 *gallons*
1.36 *acre-inches*
0.15 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 2 - 06/20/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	11.30	1.73	44.00	205.09
Fresh water	0.00	0.00	0.00	1,332.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	18.30	1.73	44.00	1,537.17
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	175.39	32.28	223.81	925.36
Nutrient balance	-157.09	-30.55	-179.81	611.81
Applied to removed ratio	0.10	0.05	0.20	1.66

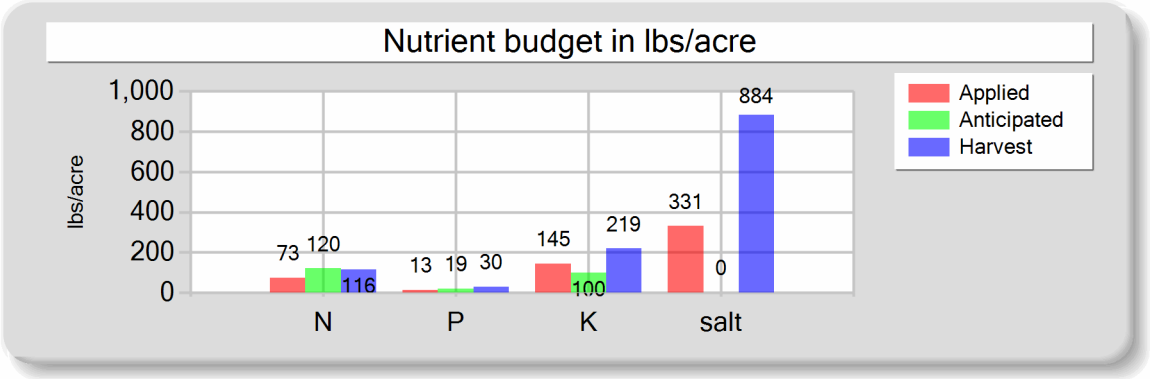
Fresh water applied	
11,051,015.00	*gallons*
406.97	*acre-inches*
45.22	*inches/acre*

Process wastewater applied	
25,931.00	*gallons*
0.95	*acre-inches*
0.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/30/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	39.04	10.13	64.13	57.47
Process wastewater	26.79	2.83	80.72	273.07
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	72.83	12.96	144.85	330.54
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	115.78	29.75	218.69	884.40
Nutrient balance	-42.94	-16.79	-73.84	-553.86
Applied to removed ratio	0.63	0.44	0.66	0.37

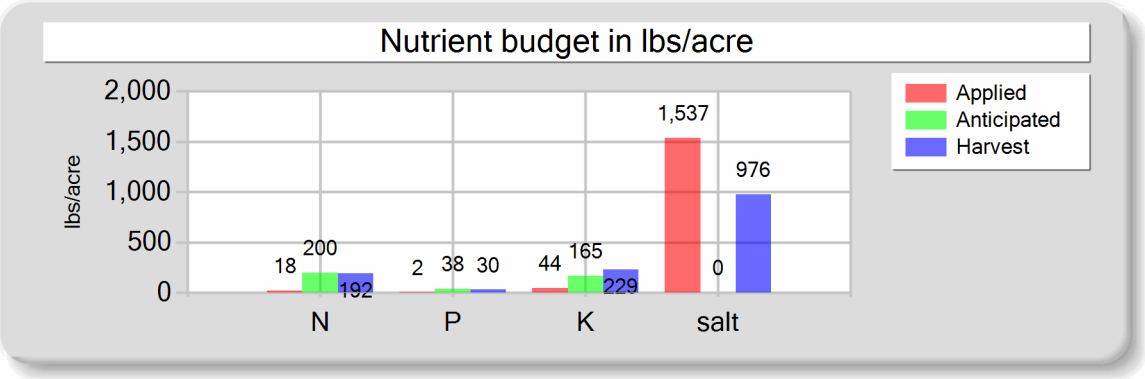
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
32,928.00	*gallons*
1.21	*acre-inches*
0.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/20/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	11.30	1.73	44.00	205.10
Fresh water	0.00	0.00	0.00	1,332.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	18.30	1.73	44.00	1,537.17
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	191.58	30.28	228.66	976.44
Nutrient balance	-173.28	-28.55	-184.66	560.73
Applied to removed ratio	0.10	0.06	0.19	1.57

Fresh water applied	
9,823,125.00	*gallons*
361.75	*acre-inches*
45.22	*inches/acre*

Process wastewater applied	
23,050.00	*gallons*
0.85	*acre-inches*
0.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Cows & Heifers

Sample and source description: Cows & Heifers

Sample date: 10/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,880.00	1,266.40	8,016.50							2.34
DL	1.00	1.00	1.00							0.01

Manure

Sample and source description: Manure

Sample date: 03/04/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	803.50	5,628.10							
DL	1.00	1.00	1.00							

Cows & Heifers

Sample and source description: Cows & Heifers

Sample date: 09/28/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 28.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	3,148.50	28,677.70							
DL	1.00	1.00	1.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 03/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	780.00		0.00	0.00	82.40	2,350.00									7,950
DL	32.00		0.10	0.40	2.50	25.00									20

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 08/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	470.00		0.00	0.00	71.90	1,830.00									8,530
DL	32.00		0.10	0.40	0.50	5.00									20

C. FRESH WATER ANALYSES

Canal Water

Aconing in

Sample description: Aconing in

Sample date: 06/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								264.00	130
DL	0.50		0.40								1.00	20

Canal Water

Field #1

Sample description: Field #1

Sample date: 06/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								262.00	140
DL	0.50		0.40								1.00	20

Field #2

Sample description: Field #2

Sample date: 06/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								261.00	130
DL	0.50		0.40								1.00	20

Field #3

Sample description: Field #3

Sample date: 06/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								260.00	130
DL	0.50		0.40								1.00	20

Domestic Well

Domestic Well

At pump

Sample description: At pump

Sample date: 09/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	4.90									
DL		0.20	0.40									

D. SOIL ANALYSES

Field 1

Field #1 Soil

Sample and source description: Field #1 Soil

Sample date: 05/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			12.00				
DL			2.00				

Field 2

Field #2 Soil

Sample and source description: Field #2 Soil

Sample date: 05/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			5.00				
DL			2.00				

Field 3

Field 3

Field #3 Soil

Sample and source description: Field #3 Soil

Sample date: 05/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			7.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Field 1 - 11/30/2022: Oats, silage-soft dough

Field 1

Sample and source description: Field 1

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	3,180.00	22,300.00		9.80
DL	500.00	2.00	20.00		0.10

Field 1 - 06/20/2023: Corn, silage

Field 1

Sample and source description: Field 1

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	2,520.00	18,600.00		7.80
DL	500.00	2.00	20.00		0.10

Field 2 - 11/30/2022: Oats, silage-soft dough

Field 2

Sample and source description: Field 2

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	3,300.00	24,000.00		10.00
DL	500.00	2.00	20.00		0.10

Field 2 - 06/20/2023: Corn, silage

Field 2

Sample and source description: Field 2

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	3,000.00	20,800.00		8.60
DL	500.00	2.00	20.00		0.10

Field 3 - 11/30/2022: Oats, silage-soft dough

Field 3

Sample and source description: Field 3

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	3,700.00	27,200.00		11.00
DL	500.00	2.00	20.00		0.10

Field 3 - 06/20/2023: Corn, silage

Field 3

Sample and source description: Field 3

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	2,450.00	18,500.00		7.90
DL	500.00	2.00	20.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

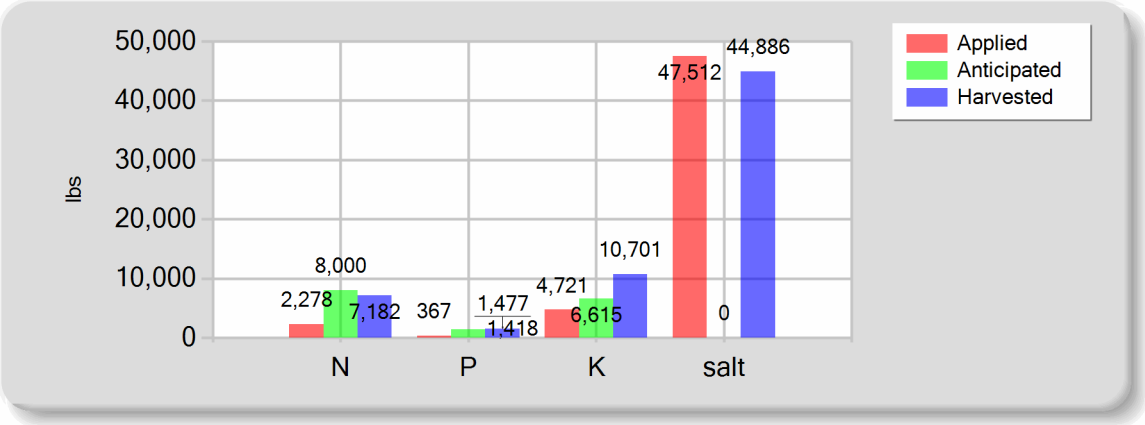
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

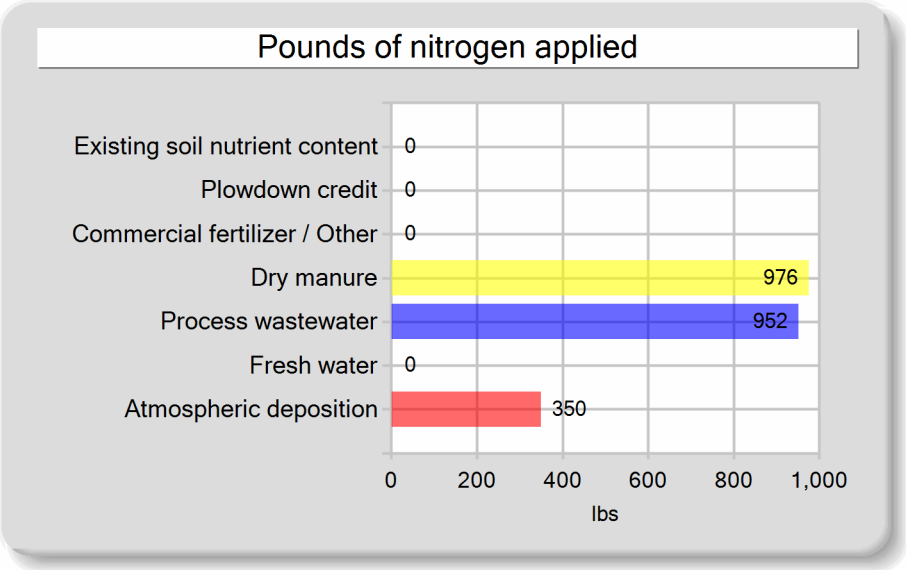
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

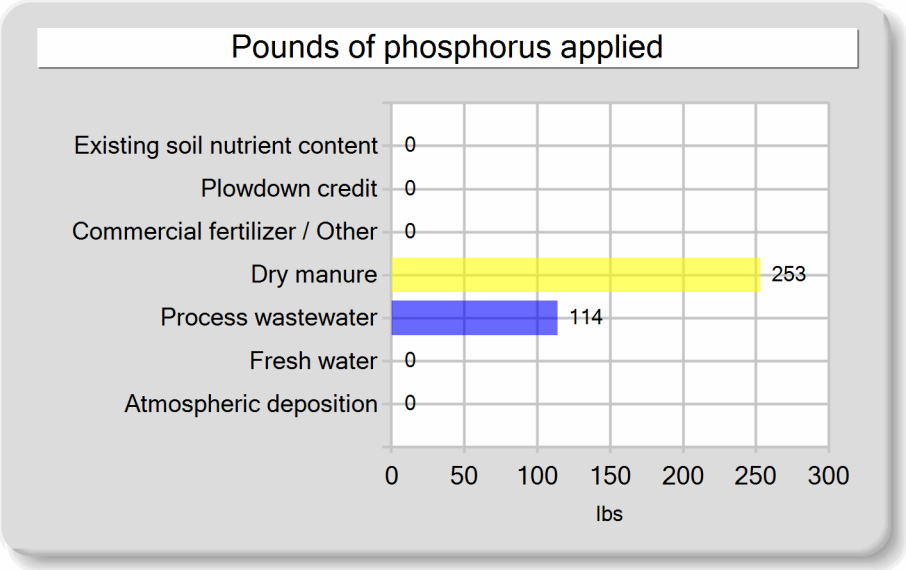
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	976.00	253.28	1,603.30	1,436.76
Process wastewater	952.30	113.98	3,117.96	11,954.04
Fresh water	0.00	0.00	0.00	34,121.67
Atmospheric deposition	350.00	0.00	0.00	0.00
Total nutrients applied	2,278.30	367.26	4,721.26	47,512.47
Anticipated crop nutrient removal	8,000.00	1,417.50	6,615.00	0.00
Actual crop nutrient removal	7,182.44	1,477.00	10,700.98	44,886.24
Nutrient balance	-4,904.14	-1,109.75	-5,979.72	2,626.23
Applied to removed ratio	0.32	0.25	0.44	1.06

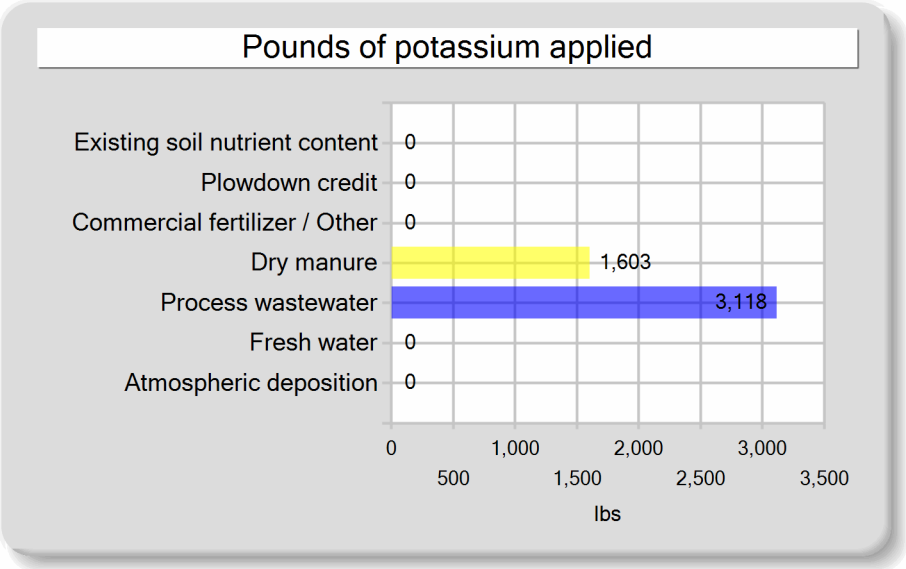
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

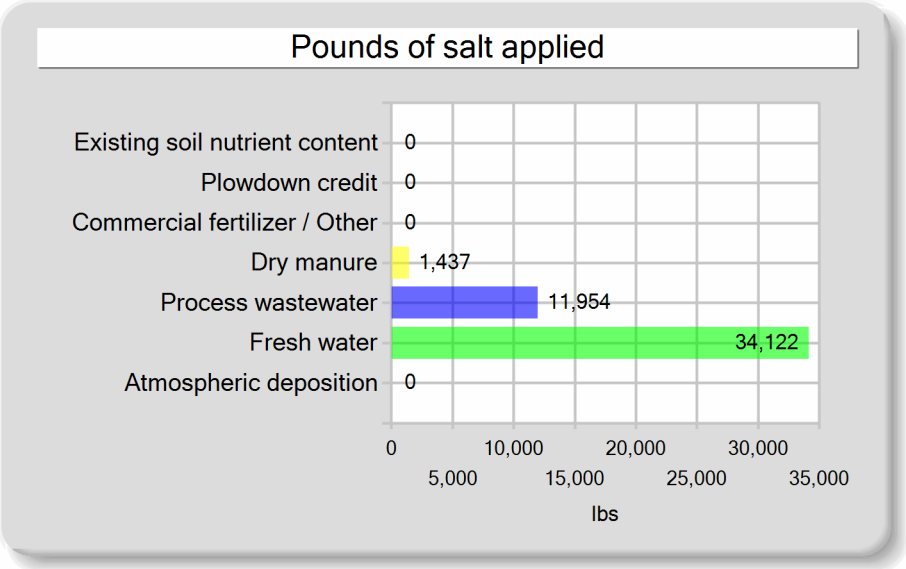


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

There were no freshwater applications for the oat crops for the 2023 Reporting Period for Field #1, Field #2, and Field #3.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jose Amaral Jose Amaral	Maria Amaral Maria Amaral
PRINT OR TYPE NAME	PRINT OR TYPE NAME
JOSE Amaral	maria Amaral
DATE	DATE
6-18-2024	6-18-2024

Sold manure
Give Away
Ester co Seco
que se dan

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: _____

Name of Dairy Facility: Amaral Dairy

Facility Address: 4715 Co Rd N Orland 95963
Number and Street — City — Zip Code

Contact Person Name and Phone Number: _____
Name — Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: K&S Spreading & Hauling, Inc.

Address of Hauling Company /Person: 4750 Clark Ave Orland, CA 95963
Number and Street — City — Zip Code

Contact Person: Luke Kampmann 530 674 2783
Name — Phone Number

Destination Information:

(Composting Facility) / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name — Number and Street — City — Zip Code — Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street — City — Zip Code — Assessor's Parcel Number

Dates Hauled: 9-28-2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 750 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 90%
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: ___0___ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: José Amaral Maria Amaral Date: 9-28-2023

Hauler's Signature: _____ Date: _____

Amaral Dairy

4715 County Road N
Orland, CA 95963

Source Description	Sample Date	Sample ID
Domestic & Agricultural Supply Wells	9/27/2023	At The Pump

October 4, 2023

Lab No. : **CH 2378410**
Customer No. : **7000172**

Joe Amaral
4715 County Road N
Orland, CA 95963

Laboratory Report

Introduction: This report package contains a total of 3 pages divided into 3 sections:

Case Narrative	(1 page)	: An overview of the work performed at FGL.
Sample Results	(1 page)	: Results for each sample submitted.
Quality Control	(1 page)	: Supporting Quality Control (QC) results.

Case Narrative

This Case Narrative pertains to the following samples:

Sample Description	Date Sampled	Date Received	FGL Lab No.	Matrix
At The Pump	09/27/2023	09/28/2023	CH 2378410-001	GW

Sampling and Receipt Information:

The Sample was received in acceptable condition and within temperature requirements, unless noted on the Condition Upon Receipt (CUR) form. The Sample was received, prepared and analyzed within the method specified holding times. All samples arrived on ice. All samples were checked for pH if acid or base preservation is required (except for VOAs). For details of sample receipt information, please see the associated Chain of Custody and Condition Upon Receipt Form.

Quality Control: All samples were prepared and analyzed according to established quality control criteria. Any exceptions are noted in the Quality Control Section of this report.

Test Summary

SM 4500-NH3 G	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 4500-NO3 F	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)

Certification: I certify that this data package is in compliance with ELAP standards, both technically and for completeness, except for any conditions listed above and in the QC Section. Release of the data contained in this data package is authorized by the Laboratory Director or his designee, as verified by the following electronic signature. This report shall not be reproduced except in full, without the written approval of the laboratory.

KD: EHB

Approved By **Kelly A. Dunnahoo, B.S.** Digitally signed by Kelly A. Dunnahoo, B.S.
Title: Laboratory Director
Date: 2023-10-04

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
Env FAX: (805)525-4172 / Ag FAX: (805)392-2063
CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
CA ELAP Certification No. 1563

Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
CA ELAP Certification No. 2670

Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
CA ELAP Certification No. 2775

Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
CA ELAP Certification No. 2810

October 4, 2023

Joe Amaral
4715 County Road N
Orland, CA 95963

Lab No. : CH 2378410-001
Customer No. : 7000172

Sampled On : September 27, 2023 at 19:00
Sampled By : Maria Amaral
Received On : September 28, 2023 at 09:42
Matrix : Ground Water

Description : At The Pump
Project : Domestic and Agricultural Supply Wells

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
							Date	Time	Who	Method	Date	Time	Who
Dairy Analysis													
Ammonia-N	ND	0.2	mg/L		1	U	10/03/2023	19:25	lcr	SM 4500-NH3 G	10/03/2023	22:18	lcr
Nitrate Nitrogen	4.9	0.4	mg/L		1		09/29/2023	12:00	lfs	SM 4500-NO3 F	09/29/2023	14:44	lfs

DQF Flags Definition:
U Constituent results were non-detect.

ND=Non-Detected, RL=Reporting Level , Dil.=Dilution

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
Env FAX: (805)525-4172 / Ag FAX: (805)392-2063
CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
CA ELAP Certification No. 1563

Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
CA ELAP Certification No. 2670

Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
CA ELAP Certification No. 2775

Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
CA ELAP Certification No. 2810

October 4, 2023
Joe Amaral

Lab No. : CH 2378410
Customer No. : 7000172

Quality Control - Wet Chem

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
Wet Chem								
Ammonia Nitrogen	4500NH3G	10/03/2023:211131LCR	Blank	mg/L		ND	<0.2	
			LCS	mg/L	2.000	92.8%	90-110	
			MS	mg/L	2.000	93.9%	46-139	
		(SP 2316353-002)	MSD	mg/L	2.000	93.2%	46-139	
			MSRPD	mg/L		0.7%	≤20	
Nitrate Nitrogen	4500NO3F	09/29/2023:210984LFS	Blank	mg/L		ND	<0.4	
			LCS	mg/L	11.22	98.2%	80-120	
			MS	mg/L	5.609	83.3%	66-125	
		(SP 2316421-002)	MSD	mg/L	5.609	82.2%	66-125	
			MSRPD	mg/L		0.2%	≤30.4	

Definition

Blank : Method Blank - Prepared to verify that the preparation process is not contributing contamination to the samples.

DQO : Data Quality Objective - This is the criteria against which the quality control data is compared.

LCS : Laboratory Control Standard/Sample - Prepared to verify that the preparation process is not affecting analyte recovery.

MS : Matrix Spikes - A random sample is spiked with a known amount of analyte. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSD : Matrix Spike Duplicate of MS/MSD pair - A random sample duplicate is spiked with a known amount of analyted. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSRPD : MS/MSD Relative Percent Difference (RPD) - The MS relative percent difference is an indication of precision for the preparation and analysis.

ND : Non-detect - Result was below the DQO listed for the analyte.

Page 3 of 3

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
Env FAX: (805)525-4172 / Ag FAX: (805)392-2063
CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
CA ELAP Certification No. 1563

Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
CA ELAP Certification No. 2670

Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
CA ELAP Certification No. 2775

Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
CA ELAP Certification No. 281

FGL ENVIRONMENTAL AGRICULTURAL
Analytical Chemists

Annual 9-27-2023

CHAIN OF CUSTODY

www.fglinc.com

Laboratory Copy (1 of 2)

72040:01/02/2023

TEST DESCRIPTION - See Reverse side for Container, Preservative and Sampling information

Corp

Client: **Joe Amaral**
Address: 4715 County Road N
Orland, CA 95963
Drinking water

Phone: (530)865-7558 Fax: (530)865-7558
Contact Person: Maria Amaral
Project Name: **Domestic & Ag Supply Wells**
Purchase Order Number:
Quote Number: **STK20160224-01** 2023

Sampler(s)
Maria Amaral

Sampling Fee: _____ Pickup Fee: _____
Compositor Setup Date: ___/___/___ Time: ___/___

Lab Number: **CH** 7378410 **7-172**

Samp Num	Location Description	Date Sampled	Time Sampled	Method of Sampling: Composite(C) Grab(G)	Type of Sample **SEE REVERSE SIDE**	Potable(P) Non-Potable(NP) Ag Water(AgW)	Bacti Type: Other(O) System(SYS) Source(SR) Waste(W)	Bacti Reason: Routine(ROUT) Repeat(RPT) Replace(RPL) Other(O) Special(SPL)	Field Test- DAW (Anually) Field E.C.	Field Test- DAW (Anually) Field NH3-N	Dairy Analysis- DAW1 (Anually) NO3-N 8oz(P)	Dairy Analysis- DAW1 (Only if detected[1] in field) NH3-N 16oz(P)-H2SO4	Sampling Directive- DAW1: Sample NO3-N each year with field tests and if field measurement indicates the presence of NH3-N then a laboratory sample of NH3-N analysis is required. See [1] in Remarks below.			
1	at the Pump	9-27-23	7:00 pm	G	GW						1	1	X			

Remarks:
Domestic and Agricultural Supply Wells
1.] If field measurement indicates the presence of ammonium nitrogen, the discharger shall collect a sample for laboratory analysis of ammonium nitrogen.

Relinquished	Date:	Time:	Relinquished	Date:	Time:	Relinquished	Date:	Time:
Maria Amaral	9-28-2023	9:42		9/28/23	1700	GLS	9/29/23	1215
Received By:	**Date:**	**Time:**	**Received By:**	**Date:**	**Time:**	**Received By:**	**Date:**	**Time:**
	9/28/23	942	GN	9/28/23	1700	MC	/	/

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
Phone: (805) 392-2000
Env Fax: (805) 525-4172 / Ag Fax: (805) 392-2063

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
Phone: (209) 942-0182
Fax: (209) 942-0423

Office & Laboratory
563 E. Lindo
Chico, CA 95926
Phone: (530) 343-5818
Fax: (530) 343-3807

Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
Phone: (805) 783-2940
Fax: (805) 783-2912

Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
Phone: (559) 734-9473
Fax: (559) 734-8435

2378410

Inter-Laboratory Condition Upon Receipt (Attach to COC)

Sample Receipt at: STK CC CH VI

1. Number of ice chests/packages received: Rec Shipping tracking # _____
2. Were samples received in a chilled condition? Temps: Rec / ____ / ____ / ____ / ____
   Surface water SWTR bact samples: A sample that has a temperature upon receipt of >10° C, whether iced or not, should be flagged unless the time since sample collection has been less than two hours.

3.	Do the number of bottles received agree with the COC?	Yes	No	N/A
4.	Were samples received intact? (i.e. no broken bottles, leaks etc.)	Yes	No	
5.	VOAs checked for Headspace?	Yes	No	N/A
6.	Were sample custody seals intact?	Yes	No	N/A
7.	If required, was sample split for pH analysis?	Yes	No	N/A
8.	Were all analyses within holding times at time of receipt?	Yes	No	
9.	Verify sample date, time and sampler name	Yes	No	

Sign and date the COC, place in a ziplock and put in the same ice chest as the samples.
Sample Receipt Review completed by (initials): CM

Sample Receipt at SP:

1. Were samples received in a chilled condition? Temps: 16 / ____ / ____ / ____ / ____
   Acceptable is above freezing to 6E C. If many packages are received at one time check for tests/H.T.'s/rushes/
2. Shipping tracking numbers: 560207914

3.	Do the number of bottles received agree with the COC?	Yes	No	N/A
4.	Were samples received intact? (i.e. no broken bottles, leaks etc.)	Yes	No	
5.	Were sample custody seals intact?	Yes	No	N/A

Sign and date the COC, obtain LIMS sample numbers, select methods/tests and print labels.

Sample Verification, Labeling and Distribution:

1.	Were all requested analyses understood and acceptable?	Yes	No		
2.	Did bottle labels correspond with the client's ID's?	Yes	No		
3.	Were all bottles requiring sample preservation properly preserved? [Exception: Oil & Grease, VOA and CrVI verified in lab]	Yes	No	N/A	FGL
4.	VOAs checked for Headspace?	Yes	No	N/A	
5.	Have rush or project due dates been checked and accepted?	Yes	No	N/A	
6.	Were all analyses within holding times at time of receipt?	Yes	No		

Attach labels to the containers and include a copy of the COC for lab delivery.
Sample Receipt, Login and Verification completed by (initials): KDC

Discrepancy Documentation:

Any items above which are "No" or do not meet specifications (i.e. temps) must be resolved.

1. Person Contacted: _____ Phone Number: _____
   Initiated By: _____ Date: _____
   Problem:
   Resolution:

2. Person Contacted: _____ Phone Number: _____
   Initiated By: _____ Date:
   Problem:
   Resolution:

(Please use the back of this sheet for additional comme
contacts)

(7000172)
Joe Amaral
CH 2378410
mdc 09/29/2023 08:41:51
CH 2378410

ghd.com

The Power of Commitment